

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that: the client (192.51.100.15) sent a DNS query (A record lookup) for yummyrecipesforme.com to the server at 203.0.113.2 over UDP port 53, and this exact same query was repeated three times at roughly two-minute intervals (13:24:32, 13:26:32, 13:28:32), all carrying the same transaction ID, indicating the resolver was retrying after receiving no usable answer.

This is based on the results of the network analysis, which show that the ICMP message returned was: a Destination Unreachable error, specifically a "port unreachable" message (ICMP type 3, code 3), not an echo reply, telling the client that nothing was listening on the destination port it had queried.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message:

The port noted in the error message is used for: DNS (53)

The most likely issue is: DNS service unavailable / port 53 closed on the target server, causing repeated failed name resolution

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: Based on the timestamps in the capture, the issue was first observed at 13:24:32 and was still occurring as of 13:28:50, the last logged retry

Explain how the IT team became aware of the incident: Several customers of clients reported that they were not able to access the client company website www.yummyrecipesforme.com, and saw the error "destination port unreachable" after waiting for the page to load.

Explain the actions taken by the IT department to investigate the incident: The team

captured live traffic between the affected client and the DNS server to observe the request/response cycle directly. They confirmed the client was correctly sending valid DNS A-record queries to 203.0.113.2 over UDP port 53. They then noted that instead of a normal DNS response, the server was returning ICMP "destination unreachable, port unreachable" messages, which shifted the investigation from "is the client misconfigured" to "is the DNS server's port 53 actually reachable/listening

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): UDP port 53 on 203.0.113.2 is closed or has nothing listening on it, since the host is actively generating ICMP type 3 / code 3 (port unreachable) responses rather than just timing out or silently dropping packets. The same query (same transaction ID) was retried three separate times roughly two minutes apart with an identical failure each time, confirming a persistent condition rather than a transient network glitch

Note a likely cause of the incident: